

Predicting Homelessness Rates From Housing and Urban Development Data

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Homelessness is one of the most harrowing socioeconomic outcomes that an individual can face, but it is also a broader societal problem influenced by local housing availability, rental costs, resources, and the availability of financial assistance. Many governmental bodies, namely the Department of Housing and Urban Development, dedicate resources to reducing homelessness rates and improving outcomes for unhoused citizens.

The objective of this analysis is to build a model that can predict homelessness rates from potential predictors such as education, employment, home availability, and cost burdens. The relevant data is provided by the U.S. Department of Housing and Urban Development from their Continuum of Care (or “CoC”) project, which attempts to track homelessness outcomes against relevant variables, and which is tracked both across regions and over time. A usefully predictive model would be of assistance to homelessness rate reduction initiatives and allow them to identify benchmarks of progress, as well as the most useful contributors to these goals.

Using Data From the Department of Housing and Urban Development

This project will use the Department of Housing and Urban Development’s Office of Policy Development and Research’s Homelessness Prediction Model Dataset (n.d.), which contains approximately 332 features with which to develop an analytical model. This dataset was collected over the course of more than a decade for a diverse set of regions, both urban and rural, and represents one of the most informationally-dense publically-available sources on the subject.

One of the major benefits of the HUD CoC dataset is its relative cleanness, completeness, and legibility. The majority of this dataset does not need any preparation, and variables, while already clearly named, are described further in the companion codebook. However, several

adjustments have been made to produce a high-quality predictive model.

Each row within this dataset represents a Continuum of Care number for a given year. Continuum of Care numbers are identifiers given to specific regions committed to reducing homelessness by the Department of Housing and Urban Development (2021). Most pertinently, the dataset provides a useful target variable in “pit_hless_pit_hud_share,” which is the rate of homeless persons per every 10,000 people on the census of a region. This is the target variable on which a predictive model will be trained.

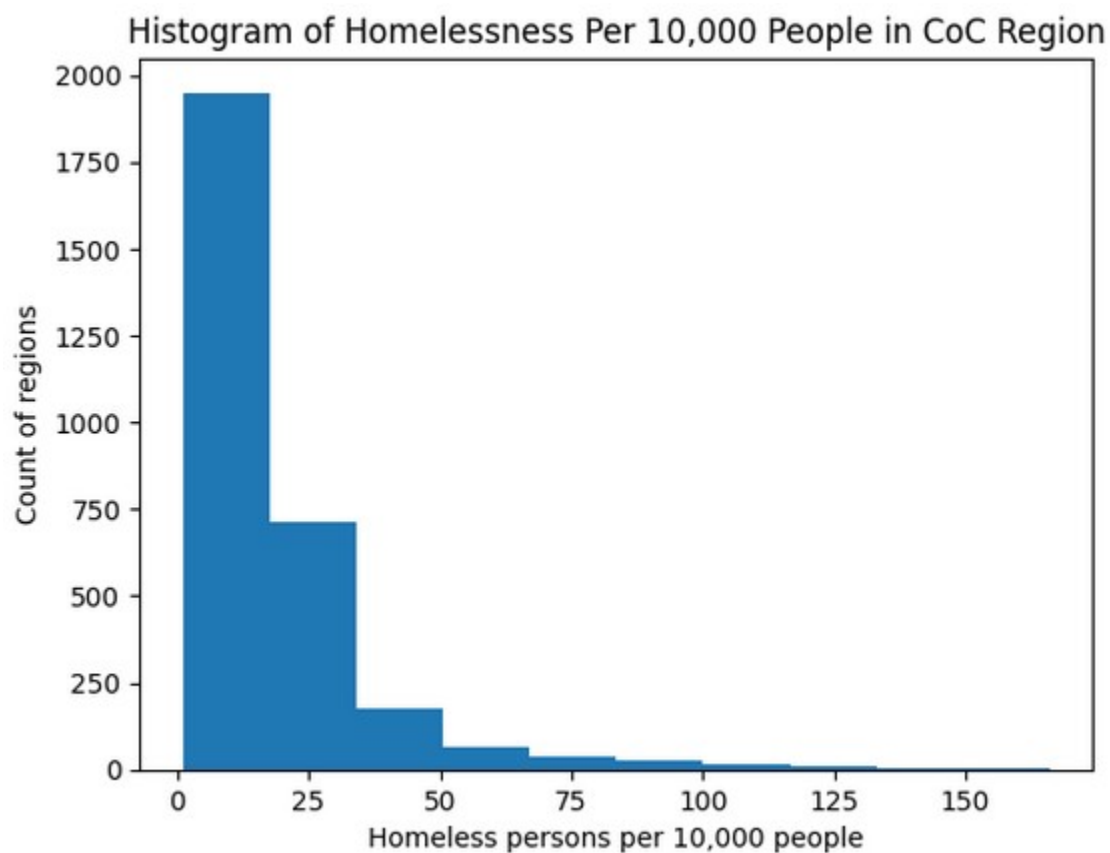
Data Preparation

After identifying the target variable, I removed all outcome-related variables, e.g. all variables listed as being HUD-related or HUD performance measures. For predictive purposes, only the target variable within this dataset should be outcome-related. This reduces the possibility of training a model on correlated outcome metrics.

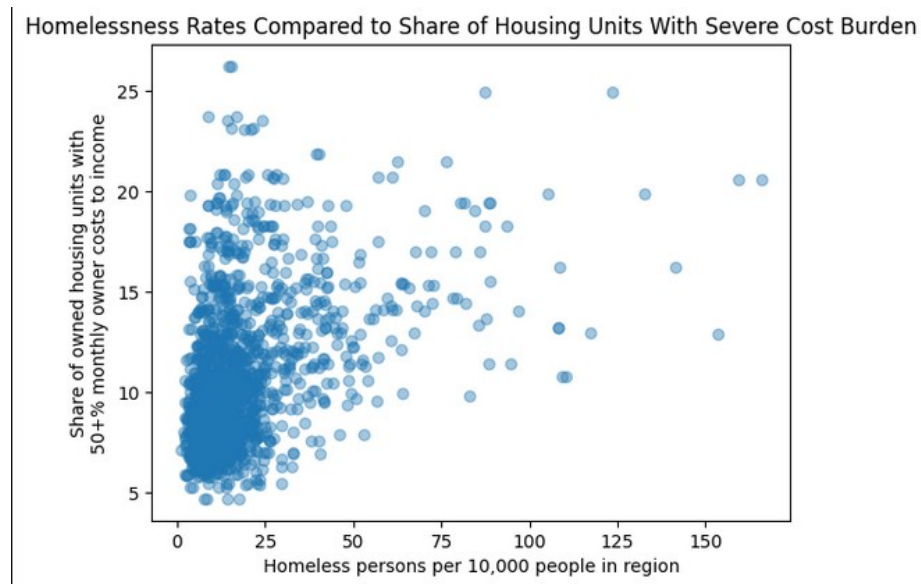
While this dataset contains time-based data, I elected not to split the dataset by time, or examine it through a time-series lens, for two reasons: Firstly, my goal is to identify predictors of homelessness from the features given in the dataset, while avoiding accounting for external features such as political shifts, e.g. those seen in 2016-2018 and onward. Therefore, even if time series data were a useful approach, my ideal model for this task could predict outcomes not because of, but indifferent to, a given year. Secondly, limiting data down to a single year severely limits the available training data (to approximately 300 entries). Therefore, the “year” feature was removed to avoid incorporating it into the model, but otherwise ignored. Similarly, the “cocnumber” feature and its associated “state_abr” feature was removed to avoid biasing the model by state or region. This helpfully reduced the dataset to purely continuous numerical features, which is necessary to use techniques like variance threshold feature reduction.

To reduce the number of features, I used a $\geq 80\%$ variance threshold to cull features with minimal variance.

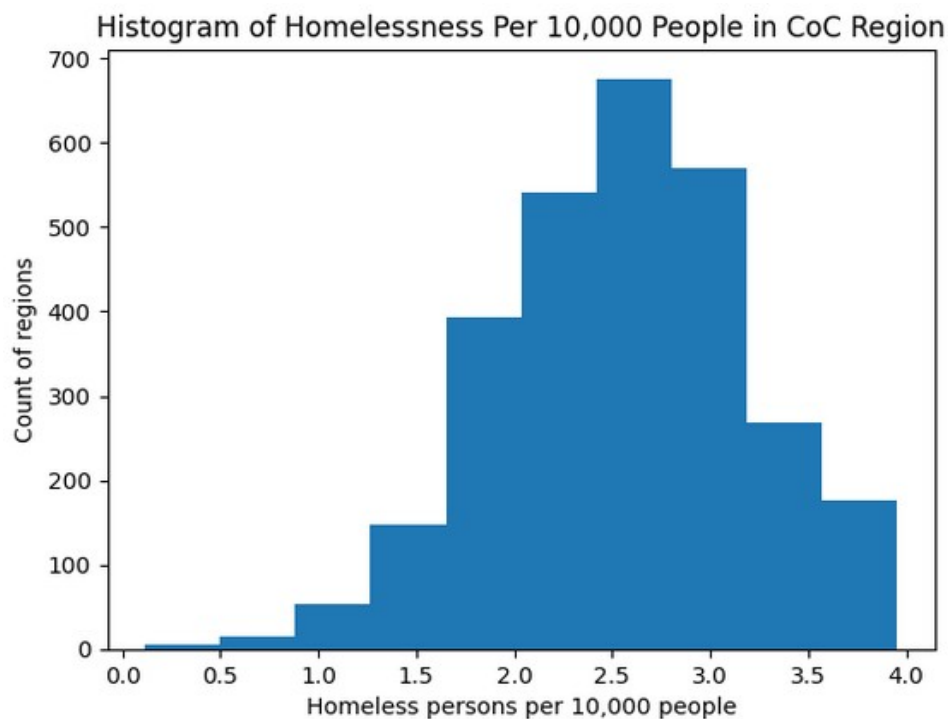
I also identified during data exploration that homelessness rates are not normally distributed. The majority of regions have low homelessness rates (at under 50 persons per 10,000 persons). This results in a fairly serious rightward skew, as seen in the homelessness rate histogram:



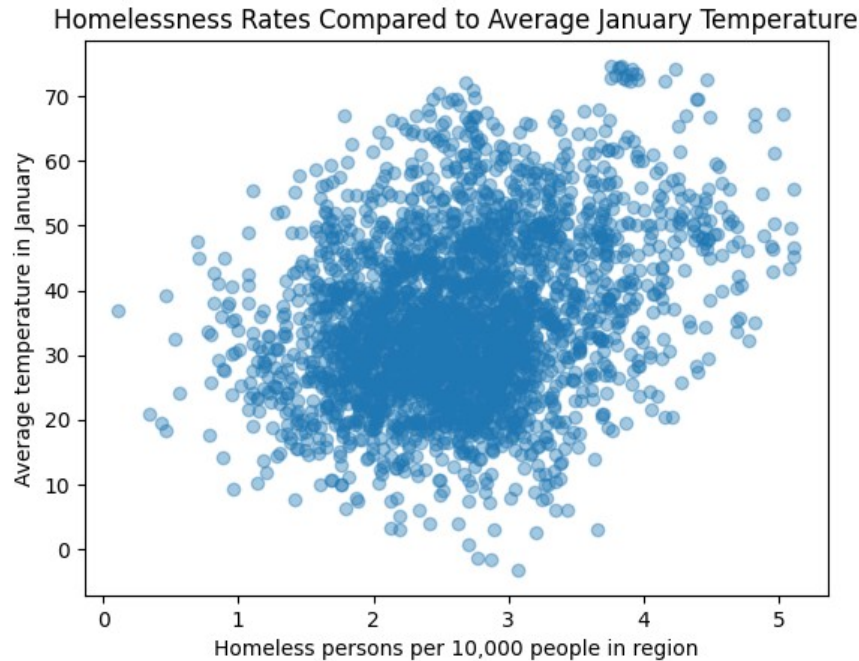
Because of this skew, many of the predictor variables are similarly skewed:



As a final preparation measure, I prepared a logarithmic homelessness rate to normalize the histogram. Making the data distribution more symmetrical is likely to result in a more accurate statistical model. The histogram of this log rate is much more normal:



This also results in the other features becoming less skewed:



Methods of Analysis

Using the normally-distributed logarithmic homelessness rate as the target variable, I first selected the 20 best features and removed redundant and unuseful features for training. I then trained and tested both a multivariate linear regression-based approach and a random forest regression-based approach.

Selecting Best Features

Before training any models, I used the univariate feature selection method `SelectKBest` (`sklearn.feature_selection.SelectKBest`) to find the twenty highest-scoring features. From these features, I removed any that seemed redundant (e.g. urbanicity score versus major city score) or that would not be useful for the model (e.g. census racial scores represent a factor that government programs cannot and should not adjust).

Multivariate Linear Regression and Ridge Regression

I initially chose a multivariate linear regression-based approach because of linear regression's utility in predicting continuous variables. I began by splitting the dataset into an

80:20 training and test data split, and began by training a simple linear regression on the training data. This simple model produced predictions with a root mean squared error (RMSE) of approximately 0.56 and an R^2 of approximately 0.26. The R^2 at 0.26 which indicates that the predictive features can account for 26% of variation; this is not ideal but is understandable for a model attempting to predict human behavior.

However, the RMSE of 0.56 is more pertinent as a metric of quality. The logarithmic homelessness rates used have a mean of 2.55 and a range of 3.84, with a minimum of 0.11, meaning that the RMSE represents a full 14% of the range.

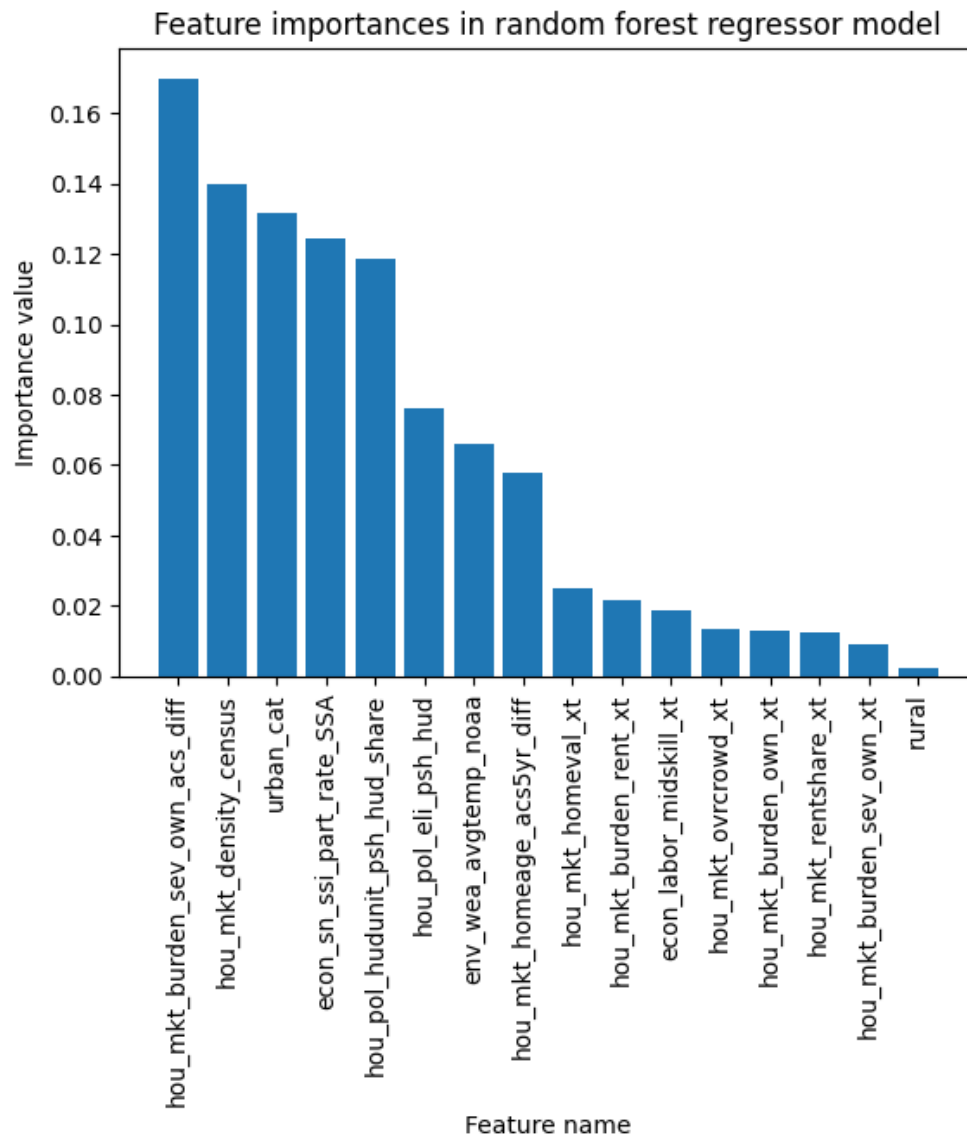
To compare model approaches, I also trained a ridge regression model on features scaled with `StandardScaler` (*sklearn.preprocessing.StandardScaler*) as preprocessing. I chose to use ridge regression to account for possible multicollinearity. I used `GridSearchCV` (*sklearn.model_selection.GridSearchCV*) to find the best-suited hyperparameters for the ridge regression and returned an alpha of 1.0, a maximum iteration count of 30, and the `sparse_cg` solver. However, this model produced a marginally *lower* R^2 and a marginally *higher* RMSE (still rounded to 0.56); it had similar but slightly worse performance.

Random Forest Regression

As an alternate approach to improve performance, I evaluated tree- and forest-based regression models. Both presented improvement, but due to the decision tree regressor models' tendency to overfit, I elected to select a random forest regressor instead. This final random forest model has an RMSE of 0.30, which is a significantly improved margin of error, and an R^2 of 0.79, which is a significantly improved predictor account of variation in values. From this model, it is possible to graph feature importances, which is instrumental in making the model findings legible to stakeholders.

Results and Findings

Using the best-predictive random forest regression model, the relative feature importances are as follows:



The most predictive feature is “hou_mkt_burden_sev_own_acs_diff,” which is a feature which (in short) represents differences over time in the proportion of homeowners for whom the home is a severe cost burden, representing $\geq 50\%$ of their income. This is an extremely interesting top predictor, because it ranks higher than the predictor which represents the severe

cost burden on its own. In short, the rapid rise in severe cost burden rates is more predictive of homelessness than the burden itself.

There are four other highly predictive features:

- “hou_mkt_density_census,” which represents housing density;
- “urban_cat,” a categorization of the urbanicity of an area;
- “econ_sn_ssi_part_rate_SSA,” representing the proportion of the population that takes SSI benefits, e.g. those who are elderly, disabled, or impoverished; and
- “hou_pol_hudunit_psh_hud_share,” which is the share of HUD-subsidized housing units to total housing units.

Conclusion

These higher-power predictors seem to indicate that the highest predictors of homelessness may be both housing availability and the impact of crisis.

Setting aside urbanicity (a factor which cannot be readily changed by the HUD or associated programs), the highest predictive factors of homelessness rates are related to the general availability of housing (density and HUD-subsidized units) and the pressures of poverty and inability to close the general population’s gap between income and housing cost. For example, homelessness within a population is correlated with receipt of social security benefits. It is important to note that many unhoused persons, without access to a permanent address, are incapable of taking social security benefits, so SSI receipt likely indicates additional impoverished members of the community alongside the unhoused. This draws a direct line between general poverty rates and general homelessness rates.

The most concerning and perhaps the most pertinent predictor is the rapid rise in the severe cost burden of homes for homeowners. More investigations are recommended for this

predictive variable, but these rates could rise due to decaying homes and infrastructure or rapid development in an area raising property costs and taxes. Both are issues which are addressable on an institutional level by budgeting property maintenance and assistance funds in impoverished areas and by extensively studying the projected impact of rapid development and gentrification in impoverished neighborhoods.

Finally, an additional next step for research would be to apply this model over time in a concentrated region to test its performance against shifting real-world metrics and refine the model. Applying this model to a time-series projection in a specific location would help illuminate the impact of actual governmental budget changes and allocations. It would particularly valuable to apply this model to at-risk regions, such as historically Black cities and neighborhoods, to identify resourcing gaps.

References

(2021, January). *The U.S. Department of Housing and Urban Development Defined CoC Names and Numbers Listed by State (January 2021)*. U.S. Department of Housing and Urban Development; U.S. Department of Housing and Urban Development._

https://www.hud.gov/sites/dfiles/CPD/documents/FY-2021_CoC-Names-Numbers_Final.pdf

Homelessness Prediction Model | HUD USER. (2025). Huduser.gov._

<https://www.huduser.gov/portal/datasets/hpmd.html>